Assignment 1 (PRACTICE)

(1) UV light of wavelength 350 nm and intensity $I = 1.0\ W/cm^2$ is directed at the Potassium surface.

(a) Find the maximum K.E. of the photoelectrons.

(b) If 0.5% of the incident photons produce photo-electrons, how many are emitted/sec. if the ~~poten~~ potassium surface has an area of $1\ cm^2$.
where work function of the Potassium

Solution Given.

$\lambda = 350\ nm.$

$I = 1\ W/m^2$

Energy of the photon

$$E_{ph} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{350 \times 10^{-9}}$$

$$E_{ph} = \frac{19.878 \times 10^{-26}}{3.5 \times 10^{-7}}$$

$$E_{ph} = 5.679 \times 10^{-19}\ \text{Joules}$$

$$E_{ph} = \frac{5.679}{1.6} = 3.5\ eV$$

$$\boxed{E_{ph} = 3.5\ eV}$$

(2)

I(c)

$$KE = E_{Ph} - \phi$$

$$KE = 3.5\,eV - 2.2\,eV$$

$$\boxed{KE = 1.3\,eV}$$

$$I = \frac{E}{A\,t}$$

$$E = I\,A\,t$$

$$= 1 \times 1 \times 10^{-4} \times 1$$

$$E = 10^{-4} \text{ Joules/sec.}$$

~~Number of photons falling on the~~. are able to

e. Number of ejected photo electrons

$$\frac{.5}{100} = 5 \times 10^{-3}$$

$$n_{Phe} = \frac{10^{-4} \times 0.005}{5.679 \times 10^{-19}}$$

$$= \frac{5}{5.679} \times 10^{12}$$

$$= 0.88 \times 10^{12}$$

$$\boxed{n_{Phe} = 8.8 \times 10^{11} \text{ photo electrons/sec.}}$$

Problem No. 2 :→ Find the photo-electric threshold for Zinc and the maximum velocity of the photoelectrons librated from its surface by electromagnetic radiation with wavelength 250 nm. $\phi = 3.74\ eV = 3.74 \times 1.6 \times 10^{-19}$ Joules

Given

$$\lambda = 250\ nm.$$

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{250 \times 10^{-9}}$$

$$E = \frac{19.878}{2.5} \times 10^{-19}$$

$$\boxed{E = 7.95 \times 10^{-19}\ \text{Joules.}}$$

Photo electric threshold ~~effect~~ wavelength.

$$\frac{hc}{\lambda} = 7.95 \times 10^{-19}$$

$$\cancel{\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{7.95 \times 10^{-19}}}$$

$$\lambda = \frac{\cancel{19.878 \times 10^{-26}}}{\cancel{7.95 \times 10^{-19}}}$$

$$\lambda = \frac{19.878 \times 10^{-26}}{3.74 \times 1.6 \times 10^{-19}}$$

$$\boxed{\lambda = \frac{19.878}{5.984} 10^{-7} = 332\ nm}$$

$$\frac{1}{2}mv^2 = E - \phi$$

$$= (7.95 - 5.984) \times 10^{-19}$$

$$\frac{1}{2}mv^2 = 1.966 \times 10^{-19}$$

$$v^2 = \frac{2 \times 1.966 \times 10^{-19}}{9.1 \times 10^{-31}}$$

$$v = \sqrt{0.432 \times 10^{+12}}$$

$$v = 0.6573 \times 10^{6}$$

$$\boxed{v = 6.5 \times 10^{5} \text{ m/sec}}$$

Problem :→ Illuminating the surface of the certain metal alternately with light wavelength $\lambda_1 = 0.35\,\mu m$ & $\lambda_2 = 0.54\,\mu m$. It was found that maximum velocity of photoelectrons differ from the factor $\eta = 2$. Find the work function of the metal.

$\lambda_1 = 0.35\,\mu m.$

$\lambda_2 = 0.54\,\mu m$

$$\frac{1}{2} m v_{1max}^2 = \frac{hc}{\lambda_1} - \phi$$

$$\frac{1}{2} m v_{2max}^2 = \frac{hc}{\lambda_2} - \phi$$

$$\frac{v_1}{v_2} = 2$$

$$\frac{v_{1max}^2}{v_{2max}^2} = \frac{\left(\frac{hc}{\lambda_1} - \phi\right)}{\left(\frac{hc}{\lambda_2} - \phi\right)}$$

$$4\frac{hc}{\lambda_2} - 4\phi = \frac{hc}{\lambda_1} - \phi$$

$$\cancel{3\phi = 4hc\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)}$$

$$\cancel{\phi = \frac{4}{3} hc\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)}$$

⑥

$$\frac{v_{1\,max}^2}{v_{2\,max}^2} = \frac{\frac{hc}{\lambda_1} - \phi}{\frac{hc}{\lambda_2} - \phi}$$

$$4\frac{hc}{\lambda_2} - 4\phi = \frac{hc}{\lambda_1} - \phi$$

$$3\phi = hc\left(\frac{4}{\lambda_2} - \frac{1}{\lambda_1}\right)$$

$$\phi = \frac{19.878 \times 10^{-26}}{3}\left[\frac{4}{0.54} - \frac{1}{0.35}\right] \times 10^{6}$$

$$= \frac{19.878 \times 10^{-20}}{3}\left[7.40 - 2.85\right]$$

$$\phi = 3.0 \times 10^{-19}$$

$$\boxed{\phi = 1.875\ eV}$$

Problem No. 10: Find the maximum K.E. and velocity of photoelectrons librated from the surface of lithium by e m γ whose electric component varies with

$\phi = 2.39\ eV$

$$E = a(1 + \cos\omega t)\cos\omega_0 t$$

where a is constant & $\omega = 6\times10^{14}\ sec^{-1}$ & $\omega_0 = 3.6\times10^{15}\ sec^{-1}$

$$E = a\cos\omega_0 t + a\cos\omega t\cos\omega_0 t$$

$$= a\cos\omega_0 t + \frac{a}{2}\cos(\omega_0 + \omega)t - a/2\cos(\omega_0 - \omega)t.$$

Maximum frequency falling on the surface.

$$\omega = 42\times10^{14}\ sec^{-1}$$

$$E = \hbar\omega = 1.05\times10^{-34}\times42\times10^{14}$$

$$= 44.1\times10^{-20}\ \text{Joules}$$

$$E = 4.41\times10^{-19}\ \text{Joules}$$

$$E = 2.75\ eV$$

$$\phi = 2.39\ eV$$

$$KE = 2.75 - 2.39 = 0.36\ eV$$

Problem No 1 A particle is defined by a wavefunction

$$\psi = A x (a^2 - x^2) \text{ for } 0 < x < a$$
$$= 0 \quad \text{for } x \leq 0 \ \& \ x \geq a$$

(i) Normalize the wavefunction.
(ii) Find the probability of finding the particle between $x_1 = a/4$ to $x_2 = \frac{3a}{4}$

Solution:

Normalization condition:

$$\int_{-\infty}^{\infty} \psi^* \psi \, dx = \int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

$$\Rightarrow \int_0^a A^2 x^2 (a^2 - x^2)^2 dx = 1$$

$$\Rightarrow A^2 \left[\int_0^a (a^4 x^2 + x^6 - 2a^2 x^4) dx \right] = 1$$

$$\Rightarrow A^2 \left\{ a^4 \cdot \frac{a^3}{3} + \frac{a^7}{7} - 2a^2 \frac{a^5}{5} \right\} = 1$$

$$\Rightarrow A^2 \left\{ \frac{a^7}{3} - \frac{9}{35} a^7 \right\} = 1$$

$$\Rightarrow A^2 a^7 \frac{8}{35} = 1$$

$$\boxed{A = \sqrt{\frac{35}{8a^7}}}$$

(ii)
$$P = \int_{a/4}^{3a/4} A^2 (a^4 x^2 + x^6 - 2a^2 x^4) dx$$

$$P = \frac{35}{8a^7} \left\{ \frac{a^4}{3} \left(\frac{27a^3}{64} - \frac{a^3}{64} \right) + \frac{a^7}{7} \left\{ \frac{2187}{16384} - \frac{1}{16384} \right\} - \frac{2a^7}{5} \left\{ \frac{243}{1024} - \frac{1}{1024} \right\} \right.$$

$$P = \frac{35}{8a^7} a^7 \left[\frac{26}{192} + \frac{2186}{7 \times 16384} - \frac{484}{5120} \right]$$

Calculate by yourself

Problem No:12: An electron is described by a wave function

$$\psi(x) = \begin{cases} 0 & \text{for } x < 0 \\ A e^{-x}(1-e^{-x}) & \text{for } x \geq 0 \end{cases}$$

(i) Normalize the wavefunction and determine A?

(ii) Find the largest probability of finding the electrons?

Solution: Applying the Normalization condition.

$$A^2 \int_0^{\infty} e^{-2x}(1-e^{-x})^2 dx = 1$$

$$A^2 \int_0^{\infty} (e^{-2x} + e^{-4x} - 2e^{-3x}) dx = 1$$

$$A^2 \left[-\frac{e^{-2x}}{2} - \frac{e^{-4x}}{4} + \frac{2}{3} e^{-3x} \right]_0^{\infty} = 1$$

$$A^2 \left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right] = 1$$

$$A^2 \frac{12+6-16}{24} = 1$$

$$\boxed{A = \sqrt{12} = 2\sqrt{3}}$$

First we will find the maximum ψ values:

$$\frac{d\psi}{dx} = 0$$

$$\Rightarrow A\{-e^{-x} + 2e^{-2x}\} = 0$$

$$\Rightarrow e^{-x}(2e^{-x} - 1) = 0$$

Either $e^{-x} = 0$

$$e^{-x} = e^{-\infty}$$

$\boxed{x = \infty}$ unacceptable because $\psi \to 0$ as $x \to \infty$

$$e^{-x} + \frac{1}{2} = 0$$

$$\Rightarrow e^{-x} = 2^{-1}$$

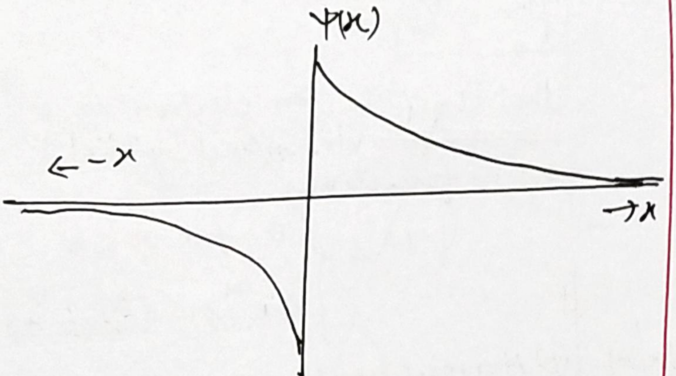
$$\Rightarrow -x \ln_e e = -1 \ln_e 2$$

$$\Rightarrow \cancel{-x = -1}$$

$$\boxed{x = \ln 2 = 0.693}$$

Problem No: 3 Check the following function can be realistic wave function of the particle

$$\psi(x) = A e^{-x/a} \quad \text{for } x > 0$$
$$= -A e^{x/a} \quad \text{for } x < 0$$



The function is finite everywhere and square integrable but it is not continuous at $x = 0$. The value of $\psi(x)$ at $x = 0$ from the expression for $x > 0$

$$\psi(0) = A$$

& for $x < 0$

$$\psi(0) = -A$$

Thus the given function does not represent ~~the does~~ ~~no~~ realistic wavefunction of the particle.

Problem No: 4

The wave function of a particle at an instant is given as.

$$\psi(x) = A e^{ikx} + B e^{-ikx} \quad \text{for } x < 0$$
$$= C e^{-kx} \quad \text{for } x > 0$$

~~Soln~~ Show that $A + B = C$

Solution

The wave function should be continuous everywhere. Thus $\psi(x)$ at $x = 0$ calculated from the first expression is equal to that from the IInd expression

$$\boxed{A + B = C}$$

Problem No. 5 Given that

$$\psi(x) = \begin{cases} A\, x/a & 0 \le x \le a \\ A \dfrac{(b-x)}{(b-a)} & a < x \le b \\ 0 & x < 0 \;\&\; x > b \end{cases}$$

(i) Normalize the wave function & find A?

(ii) Probability of finding the particle $x = 0$ to $x = a$.

Solution:→

(i) Normalization condition

$$\int_{-\infty}^{\infty} \psi^* \psi \, dx = 1$$

$$\Rightarrow \int_{-\infty}^{0} \psi^* \psi \, dx + \int_{0}^{a} \psi^* \psi \, dx + \int_{a}^{b} \psi^* \psi \, dx + \int_{b}^{\infty} \psi^* \psi \, dx = 1$$

$$\int_0^a \psi^* \psi \, dx + \int_a^b \psi^* \psi \, dx = 1$$

other integral is zero as wavefunction is zero in those ranges.

Further in this problem given wavefunction is real so

$$\psi^* = \psi$$

Now

$$A^2 \int_0^a \frac{x^2}{a^2} dx + A^2 \int_a^b \frac{(b-x)^2}{(b-a)^2} dx = 1$$

$$A^2 \left[\frac{a}{3} + \frac{1}{(b-a)^2} \int_a^b (b-x)^2 dx \, dx \right] = 1$$

Put

$b - x = t$

$-dx = dt$

x	t
a	$(b-a)$
b	0

$$A^2 \left[a/3 + \frac{1}{(b-a)^2} \int_0^{(b-a)} t^2 dt \right] = 1$$

$$\cancel{A^2 \left[a/3 + \frac{1}{(b-a)^2} \left[\frac{t^3}{3} \right] \right]}$$

$$A^2 \, a/3 + \frac{A^2}{\cancel{(b-a)^2}} \frac{1}{3} (b-a)^{\cancel{3}}] = 1$$

$$A^2 \left[\cancel{a/3} + \frac{b}{3} - \cancel{a/3} \right] = 1$$

$$\boxed{A = \sqrt{\frac{3}{b}}}$$

(ii) Probability of finding the particle in range of a.

$$P = \int_0^a \frac{3}{b} \frac{x^2}{a^2} dx$$

$$P = \frac{\cancel{3}}{b} \frac{1}{\cancel{a^2}} \frac{a^{\cancel{3}}}{\cancel{3}}$$

$$\boxed{P = a/b}$$

~~Problem 6:- An electron is described by the wavefunction~~

~~$$\psi(x) = \begin{cases} 0 & \text{for } x < 0 \\ A e^{-x} (1 - e^{-x}) & \text{for } x \geq 0 \end{cases}$$~~

~~(i) Normalize the wavefunction and determine A ?~~

~~(ii) Find largest probability of finding the electron ?~~

Problem No. 6 :→ Normalize the wavefunction and find A.

$$\psi = \begin{cases} \dfrac{A}{\sqrt{x^2+a^2}} e^{ikx} \end{cases}$$

Solution :→

Normalization condition

$$\int_{-\infty}^{\infty} \psi^* \psi \, dx = 1 \quad \text{— (1)}$$

given $\psi = \dfrac{A}{\sqrt{x^2+a^2}} e^{ikx}$

complex conjugate of the above wave ~~ψ~~ function

$$\psi^* = \frac{A}{\sqrt{x^2+a^2}} e^{-ikx}$$

Now from eq (1)

$$A^2 \int_{-\infty}^{\infty} \frac{1}{(x^2+a^2)} dx = 1$$

$$2A^2 \int_0^{\infty} \frac{dx}{x^2+a^2} dx = 1$$

$$\frac{2A^2}{a} \left[\tan^{-1} x/a\right]_0^{\infty} = 1$$

$$\frac{\cancel{2}A^2}{a} \left[\frac{\pi}{\cancel{2}} - 0\right] = 1$$

$$\boxed{A = \sqrt{a/\pi}}$$

Problem No. 7 :→ Give that

$$\psi(x) = \frac{1}{\sqrt{a}} e^{-|x|/a}$$

(i) Find probability of finding the particle ~~for~~ from $x_1 = -a$ to $x_2 = a$.

(ii) Find the value of b ~~(range x_1 (value of x_1 & x_2)~~ ($x_1 = -b$ to $x_2 = b$) so that probability of finding the particle is 0.5.

Solution :–

$$P = \int_{-a}^{a} \frac{1}{a} e^{-\frac{2|x|}{a}} dx$$

$$P = \int_{-a}^{0} \frac{1}{a} e^{\frac{2x}{a}} dx + \int_{0}^{a} \frac{1}{a} e^{-2x/a} dx$$

$$P = \frac{1}{\cancel{a}} \frac{\cancel{a}}{2} e^{2x/a}\Big|_{-a}^{0} + \frac{1}{\cancel{a}}\left(-\frac{\cancel{a}}{2}\right) e^{-2x/a}\Big|_{0}^{a}$$

$$P = \frac{1}{2}\left[1 - e^{-2}\right] - \frac{1}{2}\left[e^{-2} - 1\right]$$

$$P = \frac{1}{2}\left[1 - \frac{1}{e^2} - \frac{1}{e^2} + 1\right]$$

$$\boxed{P = \left[\frac{e^2 - 1}{e^2}\right]}$$

(ii)

$$P = \int_{-b}^{b} \frac{1}{a} e^{-\frac{2|x|}{a}} = 0.5$$

$$\Rightarrow \frac{1}{a}\left[\int_{-b}^{0} e^{+\frac{2x}{a}} dx + \int_{0}^{b} e^{-\frac{2x}{a}} dx\right] = \frac{1}{2}$$

$$\cancel{\Rightarrow \frac{1}{a}\left[\frac{a}{2} e^{\frac{2x}{a}}\Big|_{-b}^{0} + \left(-\frac{a}{2}\right) e^{-\frac{2x}{a}}\Big|_{0}^{b}\right] = \frac{1}{2}}$$

$$\Rightarrow \frac{1}{\cancel{a}}\left(\frac{\cancel{a}}{2}\right) e^{\frac{2x}{a}}\Big|_{-b}^{0} + \frac{1}{\cancel{a}}\left(-\frac{\cancel{a}}{2}\right) e^{-\frac{2x}{a}}\Big|_{0}^{b} = \frac{1}{2}$$

$$\left[\frac{1}{2}\left\{1 - e^{-\frac{2b}{a}}\right\} - \frac{1}{2} e^{-\frac{2b}{a}} + \frac{1}{2}\right] = \frac{1}{2}$$

$$1 - e^{-2b/a} = \frac{1}{2}$$

$$e^{-2b/a} = \frac{1}{2}$$

$$+\frac{2b}{a} \log_e e = + \log_e 2$$

$$b = \frac{a}{2} \frac{\log_e 2}{\log_e e}$$

$$\boxed{b = \frac{a}{2} \log_e 2}$$

Problem 8 :→ A particle limited to the x axis has the wave function $\psi(x) = ax$ between $x=0$ and $\psi = 1$ & $\psi = 0$ elsewhere.

(a) Find the probability that the particle can be found between $x_1 = 0.45$ to $x_2 = 0.55$.

(b) Find expectation value $\langle x \rangle$?

Solution

(i)

$$P = \int_{x_1}^{x} |\psi|^2 dx = \int_{0.45}^{0.55} a^2 x =$$

$$= \frac{a^2}{2}\left[(0.55)^2 - (0.45)^2\right]$$

$$= \cancel{0.0251 a^2}$$

$$\boxed{P = 0.0251 a^2}$$

If we normalize the wave function we can find a.

$$\int_0^a a^2 x^2 = 1$$

$$\frac{a^2}{3} a^3 = 1$$

$$\boxed{a = (3)^{1/5}}$$

(ii) Expectation value of x

$$\langle x \rangle = a^2 \int_0^a x^3 \, dx$$

$$\boxed{\langle x \rangle = \frac{a^5}{4}}$$